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An R Companion for Introduction to Data Mining

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Software

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Summary

An R Companion for Introduction to Data Mining is an open-source learning and teaching resource for an introductory course in data mining. It can be used to accompany the popular data mining textbook Introduction to Data Mining (Tan et al., 2017) or as a stand-alone resource to study the basic data mining concepts including data preparation, classification, clustering, and association analysis. The resource uses complete annotated examples to teach how the basic concepts are translated into R code.

The materials have been made publicly available at: https://mhahsler.github.io/Introduction_to_Data_Mining_R_Examples/book/ and licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) License.

Statement of Need

The textbook Introduction to Data Mining has been one of the most popular choices to learn data mining concepts. Several of the key chapters have been made available for free by the authors (Tan et al., 2017). One of the authors also provides Python Jupyter notebooks with examples, but complete R code examples were missing. Given the R community's interest in data analysis, data science, and machine learning, and the wide support of R packages for data mining, there was clearly a gap that needed to be filled.

During many years of teaching data mining with R, I developed the Companion for Introduction to Data Mining resource mainly based on the popular tidyverse package collection (Wickham et al., 2019), caret (Kuhn & Max, 2008), and a set of packages developed with students to better support different data mining tasks (arules (Hahsler et al., 2005), seriation (Hahsler et al., 2008) arulesViz (Hahsler, 2017), and dbscan (Hahsler et al., 2019)).

Instructional Design

The resources borrow the structure of the textbook to introduce the learner to

1. data preparation and exploratory analysis,
2. classification,
3. association rule mining, and
4. clustering.

The goal is to provide self-contained and annotated code examples that the learner can copy-and-paste into an R markdown notebook to experiment with the provided example data and then modify the code to work on their own data sets. This learning-by-doing approach has worked very well in preparing students to work with more complex real-world datasets by relieving them from the dealing with too many implementation details while exploring the concepts.

For instructors, complete presentation slide sets are provided on the book's GitHub page. The slides are organized in the same way as the companion book. A direct connection between the slides and the book is provided by the R symbol on slides where example code is available. This direct connection makes it easier for learners and instructors to switch between the concepts on the slides and the corresponding code in the companion.

The companion has been used successfully by several instructors to deliver courses in in-person and distance education settings. It is also actively maintained by faculty at the department of Computer Science at Southern Methodist University.

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